



Victoria Plaza 8th floor, 11 Parklands Road
P. O. Box 772-00606-Sarit Centre,
Westlands, Nairobi, Kenya
+254 737 666 770
info@parpus.com www.parpus.com
f d i in X PARPUS LIMITED

Retail Sales and Customer Analytics Using R

Project Title

Retail Sales Performance Analysis and Business Intelligence Using R

Project Scenario

A national supermarket chain has collected sales transactions from its branches across the country. The management has noticed fluctuations in sales, declining profits in some branches, and changing customer purchasing behavior. They have hired you as a **Junior Data Analyst** to clean, analyze, and visualize the sales data in order to identify business trends and provide recommendations that will improve sales performance and profitability.

The dataset contains **1,500+ sales transactions** collected from multiple branches and includes several data quality issues such as missing values, duplicate records, inconsistent product categories, incorrect prices, invalid dates, and formatting inconsistencies.

Section A: Data Import and Understanding

1. Import the retail sales dataset into R and verify that the dataset has been loaded successfully.
 2. Examine the structure, dimensions, variables, and data types of the retail sales dataset.
 3. Explore the dataset to identify the different categories of variables available for analysis.
 4. Produce descriptive summaries to gain an initial understanding of the dataset.
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Section B: Data Cleaning and Validation

5. Clean and validate the retail sales dataset by identifying and correcting missing values using appropriate R techniques.
6. Identify, validate, and remove duplicate transaction records while preserving valid sales information.
7. Validate customer demographic information by identifying invalid ages and inconsistent customer categories.
8. Standardize product categories, branch names, payment methods, and customer membership information.
9. Detect and correct inconsistent text formatting including unnecessary spaces, mixed letter cases, and spelling inconsistencies.
10. Convert variables into appropriate data types suitable for business analysis.
11. Identify and validate unrealistic numerical values such as negative quantities, incorrect discounts, invalid unit prices, and unrealistic profits.
12. Detect and appropriately handle outliers in sales, profit, and quantity variables.
13. Produce a cleaned and validated retail sales dataset and export it as a new CSV file.

Section C: Business Data Analysis

14. Analyze sales performance across different branches.
15. Analyze monthly and quarterly sales trends.
16. Determine the best-selling products based on quantity sold.
17. Determine the most profitable product categories.
18. Analyze customer purchasing behavior based on age and gender.
19. Compare sales generated by different payment methods.
20. Analyze customer satisfaction ratings across branches.
21. Determine the impact of promotional campaigns on sales performance.
22. Analyze the relationship between discounts and total sales.
23. Analyze return patterns and identify the most common reasons for returned products.
24. Identify branches with the highest and lowest sales performance.
25. Determine the top-performing salespersons based on total sales generated.

Section D: Data Visualization

26. Visualize total sales across branches using appropriate charts.
27. Visualize monthly sales trends using line charts.
28. Visualize product category sales using bar charts.



29. Visualize customer gender distribution.
30. Visualize customer age distribution using histograms.
31. Visualize the relationship between discounts and total sales.
32. Visualize branch profitability using boxplots.
33. Visualize customer satisfaction ratings across branches.
34. Visualize product return rates by category.
35. Produce a correlation matrix showing relationships among numerical business variables.
36. Create additional visualizations that effectively communicate key business insights obtained from the dataset.

Section E: Business Intelligence

37. Identify the factors contributing to higher sales performance.
38. Identify products with declining sales performance.
39. Determine customer segments that generate the highest revenue.
40. Rank branches according to sales performance and profitability.
41. Identify inventory management issues based on stock levels and sales patterns.
42. Generate business insights that support evidence-based decision-making.

Section F: Interpretation and Reporting

43. Interpret the results obtained from the descriptive statistical analyses.
44. Interpret each visualization and explain the key business insights.
45. Summarize the major factors influencing sales performance and profitability.
46. Develop practical recommendations for improving retail sales, customer satisfaction, and operational efficiency.
47. Prepare a comprehensive analytical report documenting the data cleaning process, methodology, analysis, visualizations, findings, conclusions, and recommendations.
48. Prepare a professional PowerPoint presentation summarizing the project methodology, key findings, visualizations, recommendations, and conclusions.
49. Present the project findings to management, clearly explaining the analytical process and supporting recommendations.